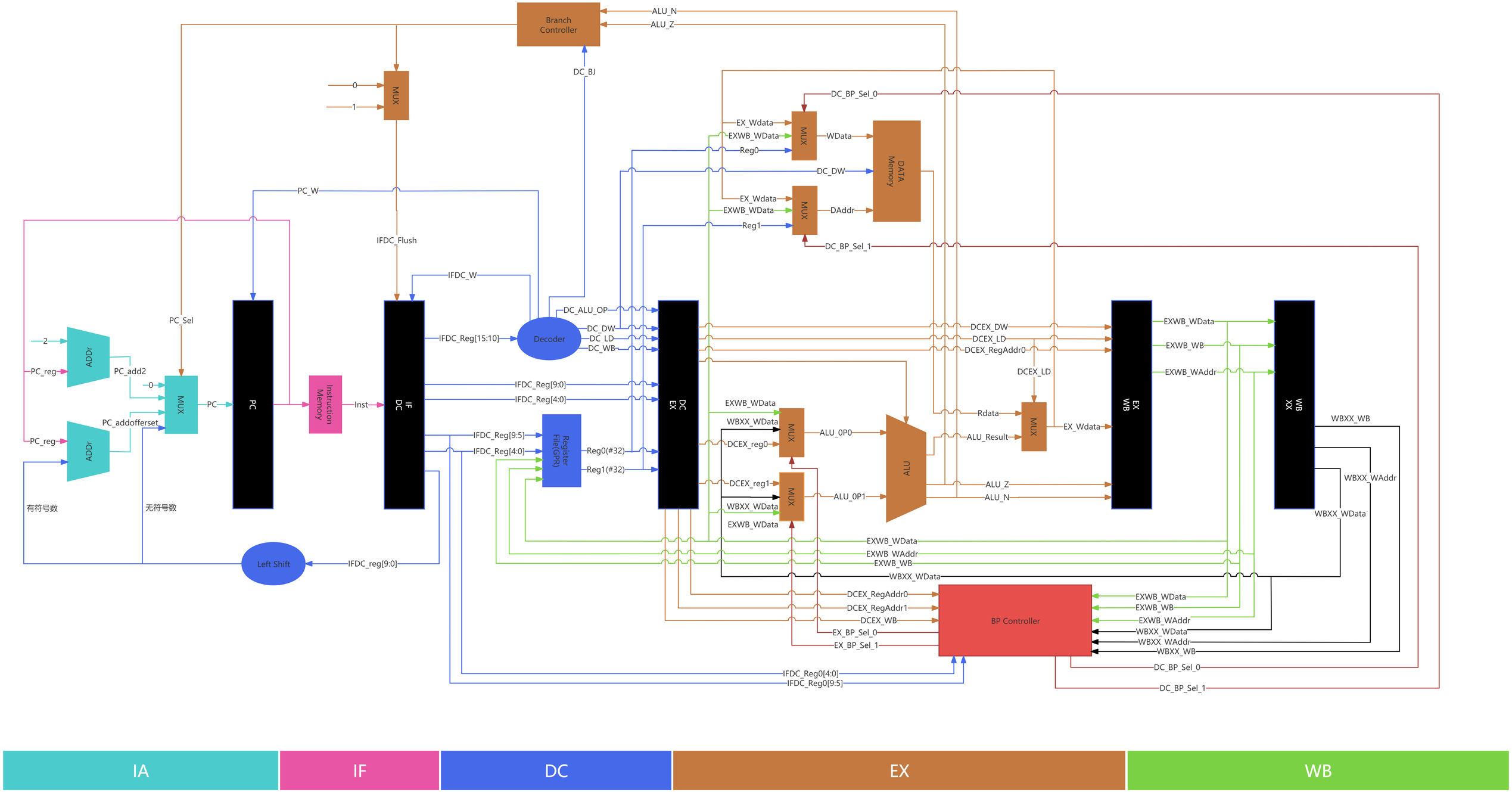
Branch Controller
ALU_N
ALU_Z
DC_BJ
MUX
0
1
PC_W
IFDC_Flush
IFDC_W
PC_Sel
2
ADDr
PC_reg
PC_add2
PC_addofferset
PC_reg
ADDr
有符号数
无符号数
MUX
PC
PC
Instruction Memory
Inst
IF DC
IFDC_Reg[15:10]
Decoder
DC_ALU_OP
DC_DW
DC_LD
DC_WB
IFDC_Reg[9:0]
IFDC_Reg[4:0]
IFDC_Reg[9:5]
IFDC_Reg[4:0]
Register File(GPR)
Reg0(#32)
Reg1(#32)
Left Shift
IFDC_reg[9:0]
DC EX
DC_BP_Sel_0
EX_Wdata
EXWB_WData
Reg0
MUX
WData
DATA Memory
DC_DW
EX_Wdata
EXWB_WData
Reg1
MUX
DAddr
DC_BP_Sel_1
EXWB_WData
WBXX_WData
DCEX_reg0
MUX
ALU_0P0
DCEX_reg1
MUX
ALU_0P1
WBXX_WData
EXWB_WData
ALU
DCEX_DW
DCEX_LD
DCEX_RegAddr0
DCEX_LD
Rdata
ALU_Result
MUX
EX_Wdata
ALU_Z
ALU_N
EX WB
EXWB_WData
EXWB_WB
EXWB_WAddr
WB XX
WBXX_WB
WBXX_WAddr
WBXX_WData
EXWB_WData
EXWB_WAddr
EXWB_WB
WBXX_WData
DCEX_RegAddr0
DCEX_RegAddr1
DCEX_WB
EX_BP_Sel_0
EX_BP_Sel_1
BP Controller
EXWB_WData
EXWB_WB
EXWB_WAddr
WBXX_WData
WBXX_WAddr
WBXX_WB
DC_BP_Sel_0
DC_BP_Sel_1
IFDC_Reg0[4:0]
IFDC_Reg0[9:5]
IA
IF
DC
EX
WB